

Venkata Krishna Varun

✉ Email

☎ (+91) 9949686580

🌐 Website

in LinkedIn

🐙 Github

Work Experience

Oracle

July 2025 – Present

Software Engineer

- Built an agentic root-cause analysis system using LLMs, RAG pipelines, and vector databases for cloud diagnostics, reducing manual debugging effort by 30%.
- Developed GenAI tooling for contextual retrieval of enterprise operational knowledge and automated distributed log analysis and failure localization.
- Engineered AI-assisted test automation covering 300+ Oracle Blockchain Platform APIs using Codex and LangChain workflows to generate tests, improve assertions, and analyze failures.
- Designed a reusable end-to-end Hyperledger Fabric API testing framework supporting request construction, response validation, transaction-state verification, and diagnostic reporting.
- Automated seven chaincode asset-lifecycle workflows (mint, transfer, approve, hold, burn, balance, history) and executed 40+ regression scenarios across 10 platform areas.

Simpl Pay

June – July 2024

Data Engineer Intern

- Replaced a daily database snapshot job with a real-time Change Data Capture (CDC) pipeline, moving data synchronization from 24-hour batches to continuous event streaming.
- Deployed Debezium on Kafka Connect with PostgreSQL source connectors to capture row-level inserts, updates, and deletes and publish them to Kafka topics.
- Delivered reliable low-latency change events to downstream systems, significantly improving data freshness and eliminating dependence on full daily snapshots.

Research Projects

Reward Decomposition in MARL — LOMAQ

Undergraduate research project; guided by Dr. Lakshmanan Kailasam

- Conducted undergraduate research on cooperative multi-agent reinforcement learning under faculty guidance, focusing on structural credit assignment and monotonic value decomposition.
- Implemented a LOMAQ-style CTDE method with per-agent utility networks and monotonic partition mixers in PyTorch; evaluated against QMIX, VDN, and IQL under matched settings.
- Designed a controlled multi-agent environment and built reproducible training/evaluation pipelines with checkpointing, deterministic execution, and automated validation.

Short-Term Residential Load Forecasting Using Deep Learning

Undergraduate research project; guided by Prof. S. K. Singh and Dr. Jayashankara M

- Investigated short-term residential electricity-load forecasting using LSTM and GRU architectures and compared them with classical regression and empirical baselines.
- Developed a leakage-aware chronological evaluation protocol with training-only preprocessing, validation-based model selection, and multiple error metrics (RMSE, MAE, R^2 , MAPE).
- Analyzed how experimental design choices (temporal splitting, preprocessing, model selection) affect reported forecasting performance and documented protocol differences transparently.

Local Credit Diagnostics

- Designed a controlled diagnostic study to examine whether low reward-reconstruction error reliably indicates high-quality cooperative decisions in multi-agent settings.
- Constructed fully enumerable synthetic cooperative games and quantified the gap between reconstruction MSE and coordination regret across additive and pairwise decompositions.
- Showed that near-zero reconstruction error can coexist with poor decision quality, highlighting limitations of standard evaluation metrics for reward decomposition.

PatchBudget

- Studied candidate verification under a fixed test-query budget using controlled QuixBugs mutations and reference oracles.
- Implemented and compared multiple test-allocation policies (round-robin, candidate-focused, failure-driven) and measured useful coverage versus false acceptance.

- Demonstrated a clear coverage–risk trade-off in budgeted verification, supporting analysis of policy behavior rather than claims of universal optimality.

Course Projects

SCLP Compiler

- Designed and implemented a compiler for a statically typed C-subset language using Flex, Bison, and C++17, generating native AArch64 and x86-64 assembly.
- Built a multi-stage pipeline with three-address code, register-transfer IR, constant propagation/folding, and target-specific code generation with stack frames and calling conventions.
- Added end-to-end compilation, execution, and rejection tests along with intermediate-representation checks.

P2P Cryptocurrency Network Simulator

- Built a discrete-event peer-to-peer cryptocurrency simulator in Python modeling network latency, transaction validation, forks, chain reorganizations, and selfish mining.
- Implemented local ledgers, mempools, message queues, and independent block-withholding strategies with reproducible seeded experiments.
- Measured propagation delay, stale blocks, throughput, tip agreement, and miner revenue across different hash-power allocations.

ChampSim Architecture Lab

- Implemented an L-TAGE-style branch predictor with geometric history lengths, tagged tables, and a loop predictor, integrated into ChampSim.
- Developed LRU, LFU, and FIFO cache replacement policies and evaluated them using deterministic synthetic workloads and IPC/MPKI metrics.
- Created reproducible experiment workflows with configuration sweeps and automated metric extraction.

xv6 Enhancements

- Extended the RISC-V xv6 operating system with demand-zero allocation, anonymous mmap, kernel threads (clone/join), and stride scheduling.
- Implemented lazy page allocation, private fork copies, memory protection, and runtime switching between round-robin and proportional-share scheduling.
- Validated all features with QEMU-based regression tests covering faults, isolation, threading, and scheduling behavior.

Turbo Compressor

- Implemented a lossless file compression utility in C++ using Huffman coding, with efficient encoding/decoding and a clean command-line interface.

Education

Indian Institute of Technology (BHU), Varanasi

Bachelor of Technology in Computer Science and Engineering

Jul 2021 - May 2025

CPI: 8.46 / 10.00

Sri Chaitanya College - Grade 11 & 12

Andhra Pradesh State Board of Intermediate Education

Jun 2019 - Jul 2021

Percentage : 96.6%

Academic Achievements & National Ranks

- **JEE Advanced 2021:** Secured **All India Rank 1353**, placing in the top **0.9%** among more than **150,000** elite candidates nationwide
- **JEE Main 2021:** Secured **All India Rank 2435**, placing in the top **0.24%** among more than **1 million** applicants nationwide
- **Quantitative Excellence:** Achieved a perfect **100% score in Mathematics** in JEE Main 2021, demonstrating strong analytical and problem-solving aptitude
- **Indian Olympiad in Chemistry (ACT):** Recognized as the **Andhra Pradesh State Topper**
- **Academic Standing:** Graduated with a **Cumulative Performance Index (CPI) of 8.46/10** in Computer Science and Engineering from **IIT (BHU) Varanasi** (Class of 2025)
- **Coursework Excellence:** Secured the highest **Grade of Excellence (A)*** in ITW1 and earned **A Grades** in core subjects including Artificial Intelligence, Data Structures, and Computer Programming

Leadership

Championship Winner | Oracle FY26 Global Tech Program

February 2026

- Spearheaded a **high-performing, 5-member team** (“Think Tank”) to architect and pitch high-impact technical solutions in a competitive, global corporate hackathon setting
- Successfully outpaced competing teams under strict delivery timelines to capture the Championship Trophy

FMC Weekend, IIT (BHU) Varanasi

Publicity Executive

- Led a cross-functional team of **25 members** to execute publicity and outreach operations for the institute’s flagship media fest
- Coordinated external outreach campaigns across **200+ colleges** to increase nationwide participation
- Contributed to digital engagement strategies resulting in **100K+ organic impressions** across social platforms

Mentoring

Telugu CSE Community, IIT (BHU)

Lead Mentor

- Mentored a cohort of **20+ students** in DSA, internship preparation, resume building, and placement strategy
- Conducted mock interviews and structured problem-solving sessions focused on software engineering recruitment
- Designed roadmap-driven preparation plans for internships and full-time technical roles

Gauth Expert

Subject Matter Expert — Advanced Mathematics

- Solved **500+** advanced mathematics problems involving Calculus, Algebra, Probability and Coordinate Geometry
- Delivered structured, step-by-step analytical explanations under strict response timelines

Sports & Extra-Curricular Activities

- Won the **Gold Medal in Boxing** at **Spardha 2022**, the annual inter-collegiate sports festival of IIT (BHU) Varanasi
- Successfully completed a **10 KM Marathon** organized for Cancer Awareness, demonstrating endurance, discipline, and commitment toward social causes

Relevant Courses

- Data Structures • Computer Programming • Object Oriented Programming • Operating Systems • Computer Networks • Database Management Systems • Artificial Intelligence • Digital Logic and Design • Computer Graphics • Computer Architecture • Probability and Statistics • Discrete Mathematics • Software Engineering • Distributed Systems • Machine Learning Fundamentals